Dr. M. Harish

Geospatial Solution Architect | Spatial AI & Earth Observation Engineer | PhD in Remote Sensing & GIS

harish.merode@gmail.com +91 994 265 8575 Chennai & Erode, Tamil Nadu, India

I architect geospatial digital twins, 3D hydrodynamic models, and Earth-observation pipelines—bridging deep satellite science with high-performance web graphics.

KEY CREDENTIALS & CERTIFICATIONS

PhD in Remote Sensing & GIS
Tamil Nadu Agricultural University (TNAU)

DGCA Commercial Drone Pilot (Rotorcraft Small)
Govt. of India · Valid through 2034

Registered Software Copyright: SW-17836/2026-CO
SWAT Weather Data Generator Platform

Sakura Science Fellow
Hirosaki University & JST, Japan

PROFESSIONAL EXPERIENCE

Senior GIS Analyst — Dronix Technologies Pvt. Ltd. (Aero360)

October 2026 – Present

Chennai, India

Leading geospatial solution architecture, satellite remote sensing intelligence pipelines, and high-performance WebGIS application engineering for enterprise and public-sector infrastructure decision-making.

- Architect end-to-end geospatial pipelines processing SAR (Sentinel-1) and optical (Sentinel-2, PlanetScope) satellite imagery for environmental monitoring and asset inspection.
- Direct the development of browser-native WebGIS platforms and 3D hydrodynamic digital twin simulations.
- Bridge field UAV photogrammetry with cloud analytics, integrating RTK drone survey data into enterprise GIS systems.
- Collaborate with multi-disciplinary engineering and remote sensing teams to deliver high-reliability spatial AI solutions.

Student cum Junior Research Fellow (JRF) — Coca-Cola Research Project

2022 – 2023

Tamil Nadu, India

Conducted satellite-based crop identification, plantation mapping, and agricultural remote sensing research combining high-resolution commercial imagery with machine learning.

- Engineered machine learning and Object-Based Image Analysis (OBIA) workflows using 3-meter PlanetScope satellite imagery.

- Executed extensive field-level ground-truthing campaigns with sub-meter RTK GPS receivers to assemble validated training and test sets.

- Achieved 87.3% classification accuracy with a 0.70 Kappa coefficient for horticultural pla… matic…
- Compiled rigorous technical reports delivering spatial acreage intelligence to agricultural project stakeholders.

Student Secretary — 10th Agricultural Students Graduate Conference (ASGC 2025)

2025

TNAU, Coimbatore, India

Elected student leader organizing an international academic conference uniting hundreds of postgraduate researchers, keynote scientists, and geospatial industry delegates.

- Coordinated peer-review and technical session tracks covering Earth observation, precision agriculture, and climate resilience.
- Facilitated keynote panels with leading scientists and senior academic faculty.
- Managed conference logistics, publication proceedings, and graduate research awards.

NCC Cadet Leader · B & C Certificate Holder — National Cadet Corps (NCC), Ministry of Defence, Govt. of India

Honors & Leadership

Tamil Nadu, India

Disciplined paramilitary training, leadership development, and marksmanship excellence.

- Awarded NCC 'B' and 'C' Certifications, representing the highest level of leadership, drill, and fieldcraft competency in the National Cadet Corps.
- Won the prestigious Shooting Gold Medal in state-level marksmanship competitions, demonstrating exceptional mental focus, breath control, and precision under pressure.
- Led contingent platoons in ceremonial parades and community service initiatives.

EDUCATION

Doctor of Philosophy (PhD) in Remote Sensing & GIS

Doctoral Degree

Tamil Nadu Agricultural University (TNAU)

Dissertation: *Assessing Impact of Climate Change on Groundwater Resources and Mapping Potential Recharge Zones using Geospatial Tools.* Coupled SWAT-MODFLOW-6, SUFI-2 calibration, and deep learning infiltration.

Master of Science (M.Sc.)

Master's Degree

Tamil Nadu Agricultural University (TNAU)

Research: *Machine Learning & Object-Based Image Analysis for Mango Plantation Mapping.* 3m PlanetScope satellite data, Random Forest classifier, 87.3% accuracy, 0.70 Kappa coefficient.

Bachelor of Science (B.Sc.)

Undergraduate Degree

Tamil Nadu Agricultural University (TNAU)

Plant tissue micropropagation, sterile culture protocols, foundational experimental design and agro-ecological science.

TECHNICAL SKILL MATRIX

Earth Observation & Remote Sensing

`Sentinel-1 SAR (C-band)` `Sentinel-2 (MSI)` `PlanetScope (3m)` `Landsat 8/9 (OLI/TIRS)` `MODIS`

Geospatial Software Engineering

`Python` `TypeScript & JavaScript` `MapLibre GL JS & Mapbox GL` `Three.js & WebGL / WebGPU`

SRTM & Copernicus DEM

React & Next.js | REST & Vector Tile APIs (MVT)

HARISH / GEOAI

Resume | Get in Touch ↗

Hydrology & Numerical Modeling

SWAT (Soil & Water Assessment Tool) | MODFLOW-6

SWAT-MODFLOW Coupled Engine

SUFI-2 / SWAT-CUP Calibration

2D Shallow Water Equations | Saint-Venant Solvers

Geospatial Data Science & Spatial AI

GeoPandas & Shapely | Rasterio & GDAL/OGR

NumPy & SciPy | Pandas | FloPy (Python MODFLOW)

Scikit-Learn (RF, SVM, XGBoost)

GIS Platforms & Photogrammetry

ArcGIS Pro & Enterprise | QGIS & GRASS GIS

Google Earth Engine (GEE) | ESA SNAP Toolbox

ERDAS Imagine | Pix4Dmapper & Pix4Dmatic

UAV & Field Intelligence

DGCA Remote Pilot License (Rotorcraft Small)

UAV Flight Planning (QGroundControl, Mission Planner)

Multispectral & RGB Payloads

RTK / PPK GNSS Base Stations

Ground Control Points (GCP) Surveying

Field Radiometry & Soil Moisture Sensors

INTELLECTUAL PROPERTY & SCHOLARLY RECORD

Global hydrological model: A comprehensive review of types, applications and uncertainties
Plant Science Today · DOI: 10.14719/pst.5960 · 2025
Authors: M. Harish, S. Pazhanivelan, M. Dhanaraju, K. Ragunath. Peer-reviewed research article in Hydrology & Physical Modeling.

Mapping of mango cropping area using machine learning techniques in Tamil Nadu
Plant Science Today · DOI: 10.14719/pst.8760 · 2025
Authors: C. Raju, S. Pazhanivelan, N. K. Sathyamoorthy, M. Harish. Peer-reviewed research article in Machine Learning & Remote Sensing.

Mango Area Mapping Using Very High-Resolution Satellite Data in Major Blocks of Krishnagiri District, Tamil Nadu, India
International Journal of Environment and Climate Change · DOI: 10.9734/ijecc/2023/v13i92302 · 2023
Authors: M. Harish, K. Ragunath, S. Pazhanivelan, C. Sivakumar, R. Kumaraperumal. Peer-reviewed research article in Satellite Earth Observation.

SWAT Weather Data Generator (SW-17836/2026-CO)
Copyright Office, Government of India · 2026
Registered proprietary software copyright for automated meteorological extraction and SWAT formatting.

Sakura Science Exchange Fellow — Weather Observation AI
Hirosaki University & JST, Japan · 2024
International scientific exchange program in weather radar, atmospheric observation, and spatial AI.

LEADERSHIP & DISTINCTIONS

- **Student Secretary:** 10th Agricultural Students Graduate Conference (ASGC 2025).
- **Shooting Gold Medalist:** State-level marksmanship competition.

- **National Cadet Corps (NCC):** 'B' and 'C' Certificate Holder, Ministry of Defence.
- **Peer Reviewer:** International Earth Observation and Remote Sensing Journals.

Dr. M. Harish

Geospatial Engineering · Earth Observation · Spatial AI

Work Research Publications About Resume Contact

Back to Top ↑

© 2026 Dr. M. Harish. All rights reserved. Registered Software Copyright: SW-17836/2026-CO.

ENGINEERED WITH NEXT.JS · TYPESCRIPT · THREE.JS · TAILWIND CSS